

ALMERIA AP, Spain

WMO: 084870

Lat: 36.846N

Lon: 2.357W

Elev: 70

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.7	45.7	19.7	14.8	52.5	24.7	18.8	54.4	33.9	59.6	30.0	58.5	10.2	340	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	93.8	69.8	90.9	70.0	87.9	70.3	78.1	83.7	76.6	82.9	75.4	82.1	19.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	138.4	81.9	74.7	130.1	80.9	73.3	123.9	80.1	41.5	83.5	40.0	83.5	38.8	82.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.1	25.7	22.7	40.0	100.1	3.0	2.6	37.9	102.0	36.1	103.5	34.5	105.0	32.3	106.9		
			33.0	79.3	3.4	2.9	30.5	81.3	28.5	83.0	26.6	84.6	24.1	86.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	55.0	55.9	59.0	62.5	67.9	74.3	79.0	80.4	75.4	69.4	61.2	56.9
	DBStd	9.67	3.19	3.52	3.40	3.76	4.34	4.87	4.44	3.93	4.00	3.94	4.11	3.29
	HDD50	9	4	3	1	0	0	0	0	0	0	0	0	1
	HDD65	1249	310	254	187	93	18	0	0	0	0	8	126	253
	CDD50	6020	159	169	281	376	553	730	899	941	762	600	337	213
	CDD65	1785	0	0	2	19	106	280	434	476	312	143	13	0
	CDH74	13257	0	0	8	81	450	1798	3848	4623	1991	444	13	1
	CDH80	4112	0	0	0	7	106	581	1299	1598	470	51	0	0
Wind	WSAvg	9.0	8.8	9.5	9.7	10.1	9.4	8.9	8.3	8.5	8.7	8.3	8.6	8.8
Precipitation	PrecAvg	7.9	0.9	0.8	0.7	0.8	0.6	0.2	0.0	0.0	0.4	1.2	1.1	1.1
	PrecMax	21.8	4.4	3.0	2.7	4.0	2.7	3.3	0.4	0.2	4.2	4.1	4.0	4.1
	PrecMin	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.4	1.0	0.7	0.7	0.8	0.7	0.6	0.1	0.1	0.7	1.1	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	69.7	74.8	81.0	87.8	95.1	97.9	97.6	90.9	84.5	75.9	69.9
		MCWB	55.8	56.4	57.0	60.4	63.5	67.9	69.0	70.0	70.7	67.0	62.7	58.5
	2%	DB	64.7	66.3	70.2	76.2	82.8	90.5	93.9	93.6	87.7	80.8	72.6	66.7
		MCWB	55.9	55.7	56.5	58.9	61.9	66.4	69.0	71.4	69.8	67.3	62.2	58.2
	5%	DB	63.0	64.4	67.8	72.0	79.0	86.3	90.3	90.9	85.0	78.3	70.1	64.9
		MCWB	55.2	55.8	56.6	58.3	61.7	66.1	70.1	71.7	70.9	67.4	61.3	57.4
	10%	DB	61.6	62.7	65.8	69.3	75.4	82.6	86.6	87.7	82.5	76.4	68.2	63.3
		MCWB	54.3	55.3	56.6	58.9	62.2	67.1	70.6	72.5	71.0	67.1	60.2	56.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	60.9	62.7	64.7	70.4	75.9	79.3	80.7	78.3	72.9	67.4	62.8
		MCDB	63.7	64.2	66.9	70.9	75.5	81.4	84.8	85.5	82.4	77.2	71.0	66.4
	2%	WB	58.6	59.2	60.8	62.9	67.7	73.5	77.3	78.6	76.1	71.1	65.1	60.7
		MCDB	62.7	63.4	65.4	69.1	73.6	79.4	83.7	84.1	81.7	76.8	69.8	64.3
	5%	WB	57.1	57.8	59.4	61.7	66.3	71.7	75.8	77.2	74.4	69.7	63.2	59.2
		MCDB	61.5	62.4	64.5	67.9	72.8	78.7	82.7	83.5	80.5	75.6	68.2	63.4
	10%	WB	55.6	56.5	58.3	60.5	65.0	70.3	74.5	76.0	72.8	68.4	61.5	57.7
		MCDB	60.3	61.3	63.7	66.9	71.7	77.4	81.5	82.7	79.5	74.5	66.7	62.1
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.4	13.6	13.3	13.3	13.7	13.1	13.0	12.9	12.7	12.8	13.3
		MCDBR	14.6	14.9	16.0	17.2	18.8	19.9	19.0	18.0	15.6	14.7	13.8	14.4
		MCWBR	11.5	10.8	10.1	9.0	9.2	9.0	9.3	9.3	8.6	9.2	9.8	10.9
	5% WB	MCDBR	14.0	13.8	14.0	13.9	13.4	13.2	12.8	12.7	12.7	12.8	12.6	13.4
		MCWBR	12.2	11.8	11.1	9.5	9.1	9.2	10.0	9.9	10.3	10.2	10.5	11.5
Clear-Sky Solar Irradiance	taub	0.322	0.351	0.395	0.397	0.421	0.449	0.461	0.462	0.429	0.399	0.354	0.329	
	taud	2.516	2.450	2.311	2.340	2.299	2.231	2.194	2.202	2.317	2.393	2.487	2.516	
	Ebn at Noon	276	280	275	281	274	266	261	258	261	259	261	265	
	Edh at Noon	25	30	38	39	42	44	46	44	38	32	26	24	
All-Sky Solar Radiation	RadAvg	860	1140	1531	1890	2203	2452	2412	2128	1686	1250	905	761	
	RadStd	70	110	121	129	156	70	48	74	85	83	72	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130
Regional (5 neighbors)	+0.58	N/A	N/A	+1.42	+0.54	N/A	N/A	N/A	+274	+189

Nomenclature: See separate page